# Analysis of parking lot occupancy

Group: CV00, Members: Kurtoglu Metehan, Sirvanci Gulce, Yakit Sezer

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## 1 Introduction

In modern surveillance systems, the use of computer vision has become essential to automate tasks that previously required manual monitoring. A practical application of these systems is parking lot management, where computer vision algorithms can detect vehicle occupancy and provide real-time data on parking space availability. This project aims to develop a computer vision system that automates parking analysis by detecting and classifying vehicles and parking spaces in real-time from surveillance camera images.

The main objective of this project is to create a system that can accurately identify parking spaces and determine their occupancy status (empty or occupied) through image analysis. In addition, the system should segment vehicles in the parking lot and classify them as correctly parked or improperly positioned outside the designated areas. To provide an intuitive and user-friendly output, the system will visualize the parking lot occupancy as a 2D top-view mini-map, where each space will be color-coded according to its status: blue for available spaces and red for occupied spaces.

The project involves processing images of an empty parking lot to determine the boundaries of individual parking spaces and then analyzing images of the parking lot with vehicles at various times of the day to assess the occupancy rate. The developed system will be evaluated on its ability to operate under different conditions, including changes in weather conditions, lighting, and the number of vehicles and pedestrians present. Despite these challenges, the system should reliably detect and classify parking spaces and ignore irrelevant elements such as pedestrians.

To evaluate the performance of the system, metrics such as the average Mean Accuracy (mAP) for parking space localization and the average Intersection over Union (mIoU) for vehicle segmentation will be used. These metrics will compare the output of the system with the actual data from the provided benchmark dataset. The dataset, which contains sequences from the PKLot dataset, contains annotations for both parking space detection and vehicle segmentation, allowing for a rigorous evaluation of the system's capabilities.

## 2 Overview

This project aimed to develop a computer vision system that automates parking analysis by detecting and classifying vehicles and parking spaces in real-time from surveillance camera images. To achieve this, the software follows these steps: First, the image was converted to grayscale, as it helps in simplifying the algorithms and also removes the complexities associated with computational requirements.

Next, the Bilateral Filter was implemented for edge preservation and noise reduction, which allows us to smooth images while preserving edges through a non-linear combination of nearby image values. This step was important as it removed most of the texture, noise and fine detail in the image, but preserved the large sharp edges without blurring them.

Then, Adaptive thresholding was implemented to adjust the value applied to the image for smaller regions, making it more flexible and effective in changing lighting conditions. The threshold dynamically for different regions of the image was calculated by this method.

The binary image was subjected to a slight Gaussian blur to minimize any remaining noise following thresholding. By adding smoothness to the image, the Gaussian blur further eliminated small noise particles that could hinder edge identification. The aim was to reduce the noise in the binary image to increase the reliability of morphological operations and line recognition operations.

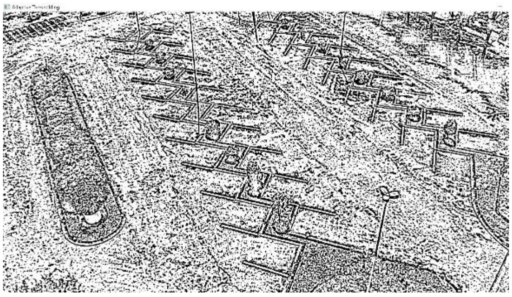
In order to improve the dependability of morphological and line recognition operations, the goal was to lower the noise in the binary image. The idea was to make the parking lines look like continuous elements in the picture by filling in the little gaps between the broken sections.

Erosion was applied to reduce the white areas (parking lines) and remove small noise elements around the lines. Erosion thinned the lines to remove smaller details. The goal was to clean up the noise by removing unnecessary small white dots that did not represent the parking lines.

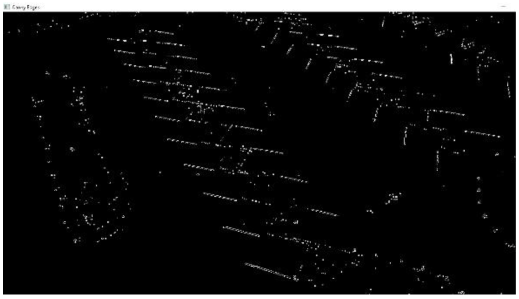
After erosion, dilation was used to thicken the parking lines once more and to muffle any small noise along the lines. The binary image's white regions (lines) were enlarged throughout this operation, bringing the parking lines back to a thickness appropriate for edge detection. By reinforcing the parking lines and expanding the white sections, the goal was to reduce any residual noise and improve line visibility in the next steps.

Canny Edge Detection was used to identify the edges of the parking lines. This technique is often used to detect high-density change areas in an image that correspond to the boundaries of objects such as parking lines. By emphasizing the edges of parking lines, it was made easier to detect them in the next line detection step.

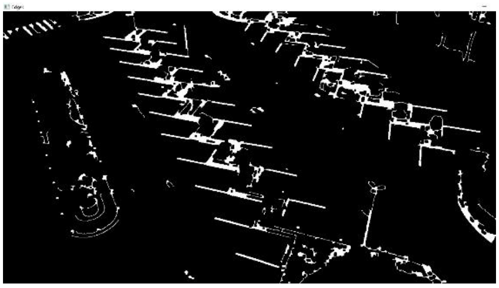
In the last step, parking lines were detected using Hough Line Transform (HoughLinesP). This algorithm detects line segments in the edge detected image. The detected lines were then drawn on a copy of the original image. The goal is to identify and draw the detected parking lines on the original image based on the edges detected in the previous step.



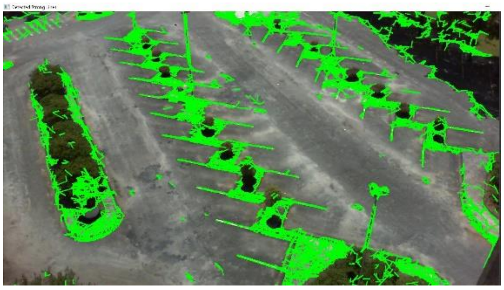
(a) Adaptive Filtering



(b) Canny Edge Detector



(c)After Preprocessing



(d) Hough Line Transform

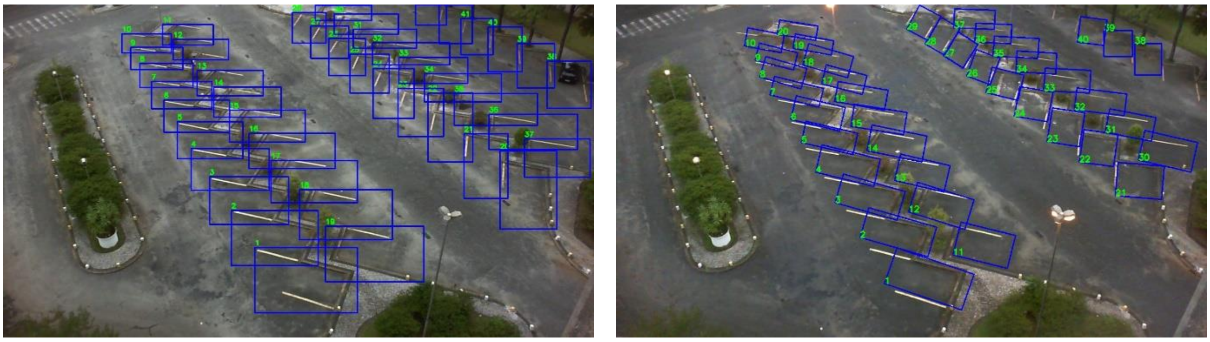
Figure 1 : Line Detection

Despite the comprehensive preprocessing steps, the desired results were not achieved primarily due to challenges posed by shadows and varying lighting conditions within the parking lot images. These environmental factors caused inconsistencies in edge detection and line recognition, leading to unreliable identification of parking spaces. Shadows often created false edges, while fluctuating lighting conditions caused the parking lines to appear fragmented or blended into the background. This significantly reduced the effectiveness of adaptive thresholding and edge detection methods, ultimately hindering the detection of parking lines with sufficient accuracy and consistency.

Given these limitations and the fixed nature of the surveillance camera, a more robust approach was required. Template Matching was identified as a suitable solution for this problem. Template Matching is a technique in computer vision used to find a portion of an image that matches a given template image. In this context, each parking space was treated as a unique template. Since the parking spaces had consistent visual characteristics when viewed from a fixed camera perspective, Template Matching was an appropriate method to reliably locate these spaces across the dataset.

To implement this approach, templates for all parking spaces were manually extracted from the initial images. Each template encapsulated the specific visual details of a parking space, including its shape, size, and any distinctive markings. This allowed for a detailed and tailored search for each parking space within the images. Using the `cv::matchTemplate` function, these templates were systematically compared against each frame in the dataset. The function slid each template over the target image, calculating a similarity measure at every location to identify regions with high correspondence to the template. Various matching methods, such as normalized cross-correlation, were explored to maximize the robustness of this detection process under different lighting conditions.

Upon identifying a match, a bounding box was drawn around the detected region. These bounding boxes were assigned unique identifiers to facilitate tracking and further analysis. Additionally, the precise location, size, and orientation of each bounding box were recorded, providing a reference framework for subsequent occupancy detection. This method demonstrated a significant improvement in accuracy over the initial preprocessing techniques. By directly matching the predefined templates to the images, the system effectively circumvented the challenges posed by shadows and lighting variations, leading to more reliable detection of parking spaces.



(a) Algorithm with bounding box (b) Ground Truth

Figure 2 : Bounding Box Comparison

The provided ground truth data, including the frames, XML files, and mask images, was loaded. These ground truth files contain detailed annotations of parking spaces and vehicle positions, serving as a benchmark for evaluating the system's accuracy.

Once the bounding boxes were established and labelled, the next step was to determine the occupancy status of each parking space. This process involved utilizing the images from the sequence0 directory, which contained five frames of the empty parking lot. These frames were used to create an average background model. First, the images underwent preprocessing steps such as grayscale conversion and noise reduction to ensure uniformity across the frames. By averaging the pixel values of these frames, a representative background image was generated, reflecting the parking lot's appearance in an unoccupied state.

This average background served as a baseline for identifying changes in the parking lot over time. Each subsequent image in the dataset was processed by subtracting this average background. The subtraction process highlighted the differences between the current frame and the background, effectively isolating potential vehicles as areas of change. However, this difference image often contained residual noise and artifacts resulting from minor fluctuations in lighting or shadows. Therefore, additional preprocessing steps, including Gaussian blurring and adaptive thresholding, were applied to the difference images. These steps further refined the results, converting the images into a binary format where potential vehicle regions were distinctly separated from the background.

The binary images were then analyzed in conjunction with the previously defined bounding boxes. For each parking space, the binary images were superimposed, and the pixel intensity within the bounding box was evaluated across all frames. This stacking approach allowed for a temporal analysis, revealing consistent changes that indicated occupancy. A threshold was applied to determine if the intensity within a bounding box was sufficient to classify the space as occupied. If the intensity exceeded this threshold, the system marked the parking space as occupied by changing the bounding box color to red. This visual cue provided an immediate and clear indication of the space's status.

# 3 Methodology

This section details the methodology employed for developing an automated parking lot management system. The methodology is divided into three key components: Car Segmentation, Parking Space Localization, and Parking Analysis. Each component is integral to accurately identifying, localizing, and assessing the occupancy of parking spaces in real-time from surveillance camera images.

## 3.1 Car Segmentation

Car segmentation is a critical step in identifying vehicles within the parking lot. The main objective here is to distinguish cars from the parking lot background while accounting for variations in lighting, shadows, and environmental conditions. This process involves several sub-steps:

**1. Background Subtraction:**

To isolate vehicles, an average background model is first created using a set of images captured when the parking lot is empty (from the sequence0 directory). This background model represents the parking lot without any vehicles, serving as a baseline for comparison.

For each image in the sequence with vehicles, background subtraction is performed by pixel-wise subtraction of the background model from the current frame. This results in a "difference image," where areas of change (i.e., vehicles) are highlighted. However, this difference image also includes noise and artifacts caused by minor variations in lighting and shadows.

**2. Preprocessing:**

**Noise Reduction:** To enhance the difference image and suppress irrelevant noise, Gaussian blurring is applied. Gaussian blur smooths the image by averaging pixel values in a local neighborhood, which helps in reducing high-frequency noise while preserving essential edges that correspond to vehicle boundaries.

**Thresholding:** Adaptive thresholding is used to convert the difference image into a binary format. This method dynamically determines threshold values for different regions of the image, making it effective in dealing with varying lighting conditions. The result is a binary image where potential vehicle regions are distinctly separated from the background.

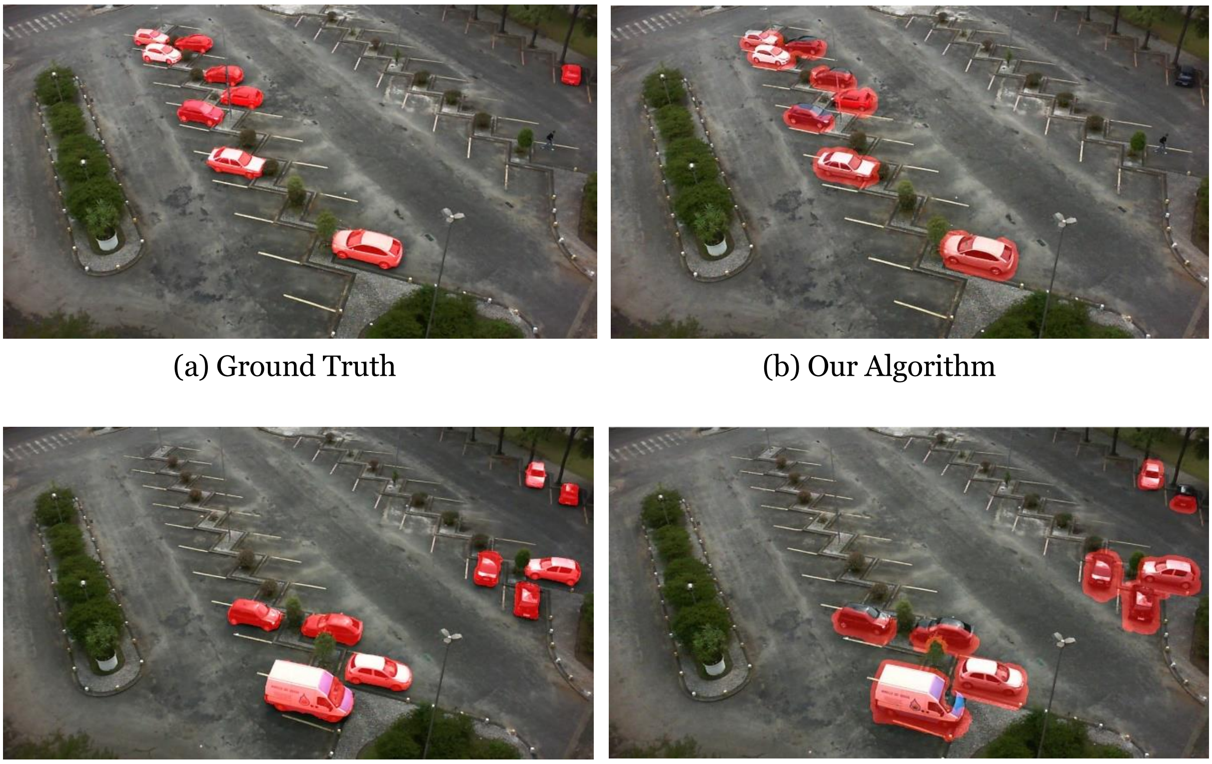
**Morphological Operations:** To further refine the binary image, morphological operations such as erosion and dilation are applied. Erosion reduces small noise elements and isolates distinct vehicle regions by shrinking the white areas (vehicles) in the image. This is followed by dilation, which enlarges the remaining regions to reconnect fragmented vehicle parts, thereby enhancing the overall shape of detected vehicles.

### 3. Contour Detection and Filtering:

With the preprocessed binary image, contour detection is performed to identify the outlines of potential objects in the scene. The contours are then analyzed based on size, shape, and aspect ratio to filter out irrelevant elements such as pedestrians or small artifacts that do not represent vehicles.

The remaining contours are presumed to correspond to vehicles. These contours are enclosed within bounding boxes, providing initial estimates of the vehicle locations within the parking lot.

Presented below are examples of the ground truth data alongside the results of our own segmentation algorithm.



(a) Ground Truth (b) Our Algorithm

(c) Ground Truth (d) Our Algorithm

Figure 3: Car Segmentation Comparison

### 4. Bounding Box Refinement:

To ensure that the bounding boxes accurately represent the vehicles, a refinement step is employed. This involves analyzing the orientation and dimensions of the bounding boxes. Rotated bounding boxes are used to better fit vehicles that are parked at an angle, ensuring that the boxes align closely with the vehicles' outlines. This refinement is crucial for subsequent steps where occupancy and positioning within parking spaces are evaluated.

## 3.2 Parking Space Localization

Localization of parking spaces is essential for determining the occupancy status of each space. This step focuses on accurately identifying the boundaries of each parking space and aligning them with the detected vehicles:

#### 1. Template Matching:

Given that parking spaces have consistent visual characteristics when viewed from a fixed surveillance camera, template matching is used to locate each parking space. Templates of all parking spaces are manually extracted from the initial empty parking lot images.

These templates are used as reference models for locating parking spaces in subsequent images. The `cv::matchTemplate` function slides each template across the target images, calculating a similarity measure (such as normalized cross-correlation) at every location. High correlation values indicate a match, allowing the system to pinpoint the exact locations of parking spaces. By using template matching, the system can effectively recognize parking spaces under varying conditions, such as changes in lighting or partial occlusion by vehicles.

#### 2. Rotated Bounding Boxes for Parking Spaces:

To accurately represent the orientation of each parking space, rotated bounding boxes are utilized. These boxes are not constrained to be axis-aligned, enabling them to match the angle and shape of the parking spaces as seen from the camera's perspective.

The orientation and size of each bounding box are adjusted based on the template matching results, ensuring that the boxes encompass the entire parking space. This precise localization is crucial for determining whether a vehicle is correctly parked within its designated space.

#### 3. Assigning Unique Identifiers:

Each parking space is assigned a unique identifier (ID), facilitating tracking and analysis. These IDs allow the system to monitor the occupancy status of each space across different frames. The IDs are stored along with the coordinates and dimensions of their corresponding bounding boxes, providing a reference framework for further processing.

### 3.3 Parking Analysis

Parking analysis involves determining the occupancy status of each parking space and evaluating vehicle positioning. This component integrates the results from car segmentation and parking space localization to provide a comprehensive analysis of the parking lot.

#### 1. Occupancy Detection:

For each localized parking space, the system analyzes the intersection between the binary images obtained from the car segmentation step and the parking space bounding boxes. This intersection highlights regions within each parking space where a vehicle is present.

The system calculates the pixel intensity within each bounding box. A threshold is applied to this intensity to classify the space as either occupied or unoccupied. If the intensity exceeds the threshold, indicating a significant presence of a vehicle, the space is marked as occupied by changing the bounding box color to red. If the intensity is below the threshold, the space is marked as unoccupied and colored blue.

This method allows the system to dynamically assess the occupancy status of each space in real-time, accommodating changes in vehicle size, positioning, and lighting conditions.

### 2. Vehicle Positioning Evaluation:

Beyond merely detecting occupancy, the system also evaluates whether vehicles are parked correctly within their designated spaces. This is achieved by comparing the vehicle's bounding box with the parking space's boundaries.

The system checks for overlap, alignment, and orientation of the vehicle's bounding box relative to the parking space. If the vehicle extends beyond the parking space's boundaries or is misaligned, it is classified as improperly parked. This information can be used for further analysis, such as identifying common parking violations or improving parking space design.

### 3. Visualization and Output Generation:

To provide an intuitive output for end-users, the system generates a 2D top-view mini-map of the parking lot. This map visualizes each parking space, color-coded based on its occupancy status: blue for available spaces and red for occupied spaces. In addition, each parking space is annotated with its unique ID and occupancy status.

The results are saved for further analysis, with detailed information such as the coordinates, dimensions, and occupancy status of each parking space stored in an easily accessible format (e.g., XML or CSV files). This enables seamless integration with external systems, such as parking management software or real-time monitoring dashboards.

### 4. Performance Metrics and Evaluation:

The system's performance is evaluated using metrics such as Mean Average Precision (mAP) for parking space localization and Mean Intersection over Union (mIoU) for vehicle segmentation. mAP measures the system's ability to accurately identify parking spaces, while mIoU assesses the overlap between predicted vehicle segmentation and ground truth data.

These metrics provide quantitative feedback on the system's accuracy and robustness, guiding further improvements in the algorithm. The system's ability to handle various environmental conditions, such as different weather and lighting scenarios, is also evaluated to ensure reliable operation in real-world settings.

# 4 Results and Discussion

In this section, we present the outcomes of our parking lot management system, detailing the steps undertaken to achieve vehicle segmentation, localization, and occupancy analysis. We outline both the successes of our implementation and the challenges encountered, particularly concerning the use of performance metrics such as mean Average Precision (mAP) and mean Intersection over Union (mIoU).

## 4.1 Ground Truth Implementation

**Loading Ground Truth Data:** We started by loading the provided ground truth data, which included frames, XML files, and mask images. These files contained detailed annotations of parking spaces and vehicle positions, serving as a benchmark for assessing the accuracy of our system.

**Drawing Bounding Boxes:** Using the information from the XML files, we drew bounding boxes around each parking space in frames from sequence0 to sequence5. Each bounding box was color-coded based on occupancy status: red for occupied spaces and blue for unoccupied spaces. This provided a visual representation of the parking lot's status according to the ground truth data.

The resulting images with annotated bounding boxes were saved in their respective directories, providing a reference point for comparing the outcomes of our custom algorithm.

**Segmentation with Masks:** We utilized the provided mask images to perform segmentation, identifying vehicles parked within the designated parking spaces. Based on these segmentation results, we further classified vehicles into two categories:

**Correctly Parked:** Vehicles parked correctly within their designated spaces were colored red.

**Incorrectly Parked:** Vehicles parked outside their designated spaces were colored green.

These segmented images were saved to the corresponding directories, completing the ground truth processing phase.

**Transition to the Custom Algorithm:** After processing and visualizing the ground truth data, we transitioned to implementing our own algorithm for parking space detection and analysis.

## 4.2 Custom Algorithm Implementation

**Parking Space Detection using Template Matching:** For sequence0, we applied template matching to detect parking spaces. For each identified parking space, we recorded the bounding box parameters, including coordinates (x, y), size, and angle.

These bounding boxes were then applied to all frames from sequence1 to sequence5, ensuring the bounding boxes accurately represented the detected parking spaces' positions and orientations in the images.

**Saving Detected Parking Spaces:** The frames with the detected parking spaces were annotated and saved in the directory. This step provided a baseline for evaluating the accuracy of our algorithm in identifying and localizing parking spaces compared to the ground truth.

**Background Subtraction for Vehicle Detection:** An average background model was created using the five empty parking lot images from sequence0. This background model served as a reference to detect changes in subsequent images.

Using this background, we subtracted it from the frames in sequence1 to sequence5 to isolate regions where vehicles were present.

After obtaining the differences, preprocessing techniques such as Gaussian blurring and adaptive thresholding were applied to produce binary images that highlight potential vehicles.

**Processing and Overlaying Results:** The binary images resulting from the preprocessing step were saved in the directory. These binary images were then overlaid onto their corresponding frames from the directory.

For each overlay, we counted the white pixels within each bounding box to determine vehicle presence. This comparison was made for each frame pair (e.g., image1 with image1, image2 with image2) to maintain consistency. However, this step presented some uncertainty, particularly regarding the matching process and pixel counting accuracy.

**Occupancy Classification:** If the number of white pixels within a bounding box exceeded a predefined threshold, the system classified the parking space as occupied and changed the bounding box color to red. Otherwise, the box remained blue.

The final images with occupancy status annotations were saved in the directory.

### 4.3 Analysis with mAP and mIoU Metrics

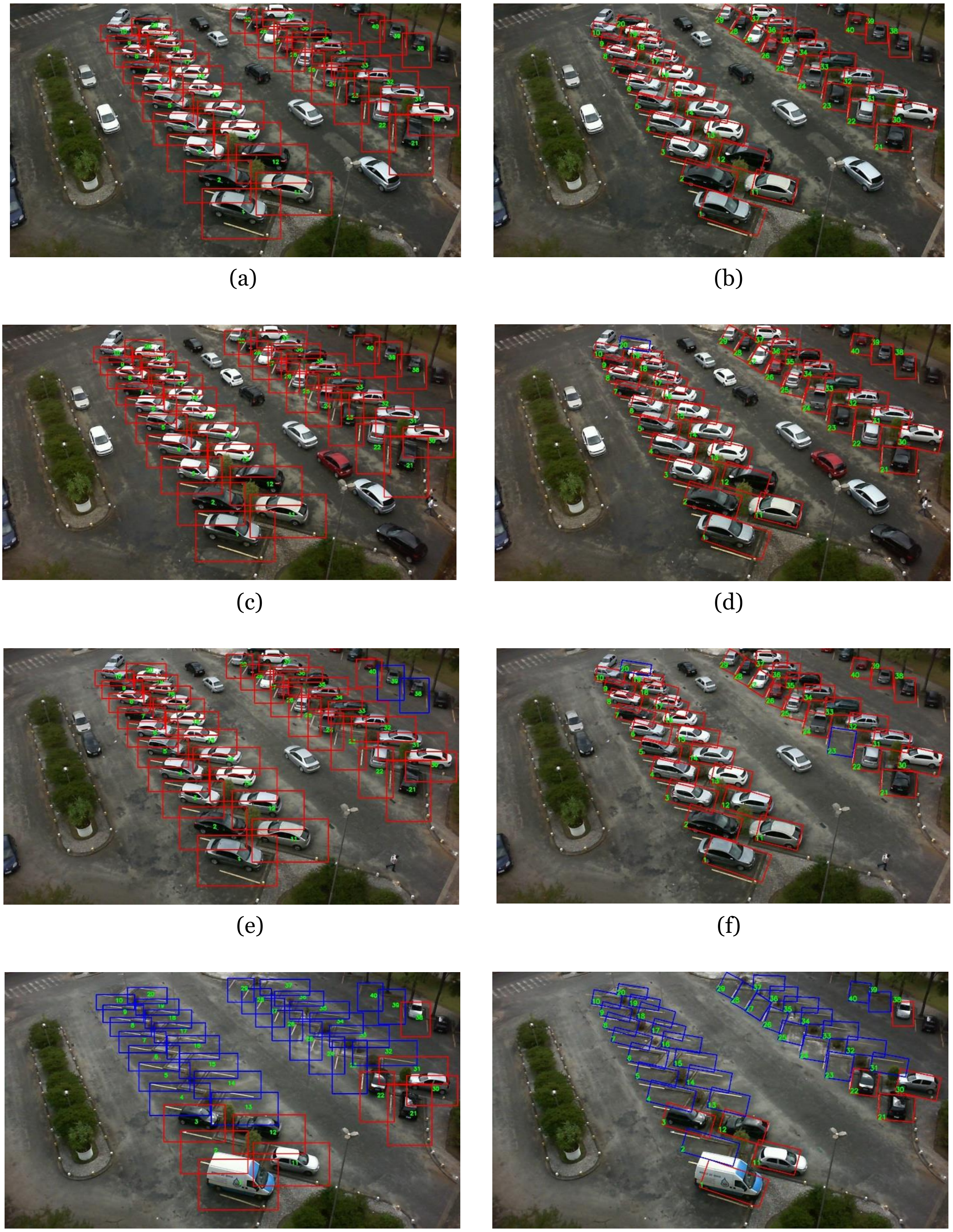
**Evaluation with mAP and mIoU:** Following the implementation of our algorithm, we intended to analyze the results using mAP and mIoU metrics. mAP was used to measure the system's ability to accurately localize parking spaces, while mIoU evaluated the overlap between our segmented vehicles and the ground truth.

### 4.4 Summary of Outcomes

Successfully loaded and visualized ground truth data, accurately drawing bounding boxes and performing segmentation based on the provided masks.

Effectively utilized template matching to identify parking spaces in sequence0, consistently applying these detections across other sequences.

Implemented background subtraction and subsequent binary image processing to detect vehicle presence, resulting in visually intuitive outcomes when overlaying these findings on the images.



(a) (b) (c) (d) (e) (f) (g) (h)

Figure 4: Car Occupancy Detection in the Parking Lot (a,c,e,g) and Ground Truth (b,d,f,h)

## 5 Challenges and Future Work

While the system produced visually acceptable results, discrepancies in quantitative evaluation using mAP and mIoU metrics were observed, likely due to implementation issues and processing limitations. Sensitivity in counting white pixels within bounding boxes, affected by noise, shadows, and inaccuracies, led to misclassifications of occupancy status. The relatively small dataset, with varying lighting conditions and shadow patterns, further complicated line detection and affected overall accuracy. Future work should focus on expanding the dataset to improve generalization, enhancing the preprocessing pipeline to handle lighting variability and shadows more effectively, and refining occupancy detection methods to reduce noise impact. Improving mAP and mIoU implementation and exploring advanced image processing techniques can contribute to a more robust and accurate system.

## 6 Conclusions

In conclusion, the developed computer vision system for parking lot management has shown promising results in automating the process of vehicle detection, parking space localization, and occupancy analysis. The system effectively identifies and localizes parking spaces within surveillance camera images, segments vehicles, and determines the occupancy status of each space. This was achieved through a combination of techniques, including template matching for parking space localization, background subtraction for vehicle detection, and binary image processing for occupancy classification.

The system's ability to accurately detect and classify parking spaces provides valuable insights into parking lot usage, enabling real-time monitoring and efficient parking management. This can help optimize space utilization, reduce congestion, and improve the overall parking experience. However, the system is not without its challenges. The accuracy of the system heavily depends on the effectiveness of preprocessing steps, such as noise reduction and adaptive thresholding, as well as the precision of the template matching process. Additionally, the system's performance can be affected by varying lighting conditions, shadows, and environmental factors present in the parking lot images.

Despite these challenges, the potential of this system in contributing to efficient parking lot management is evident. It lays the foundation for more advanced parking solutions, offering a scalable approach that can be adapted to different parking environments. Throughout the project, our team effectively managed and coordinated all phases through regular meetings, primarily conducted via Google Meet, ensuring collaborative problem-solving and decision-making.

| **name** | **student ID** | **hour** | **parts** |
|---|---|---|---|
| Metehan | 2005942 | 60 | Vehicle Detection and Segmentation, Template Matching |
| Gulce | 2085153 | 40 | Parking Space Segmentation and Localization |
| Sezer | 2071298 | 40 | Parking Space Localization and Occupancy Detection |